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ANNalog: an Artificial Neural Network (ANN) for generation of medchem analogs

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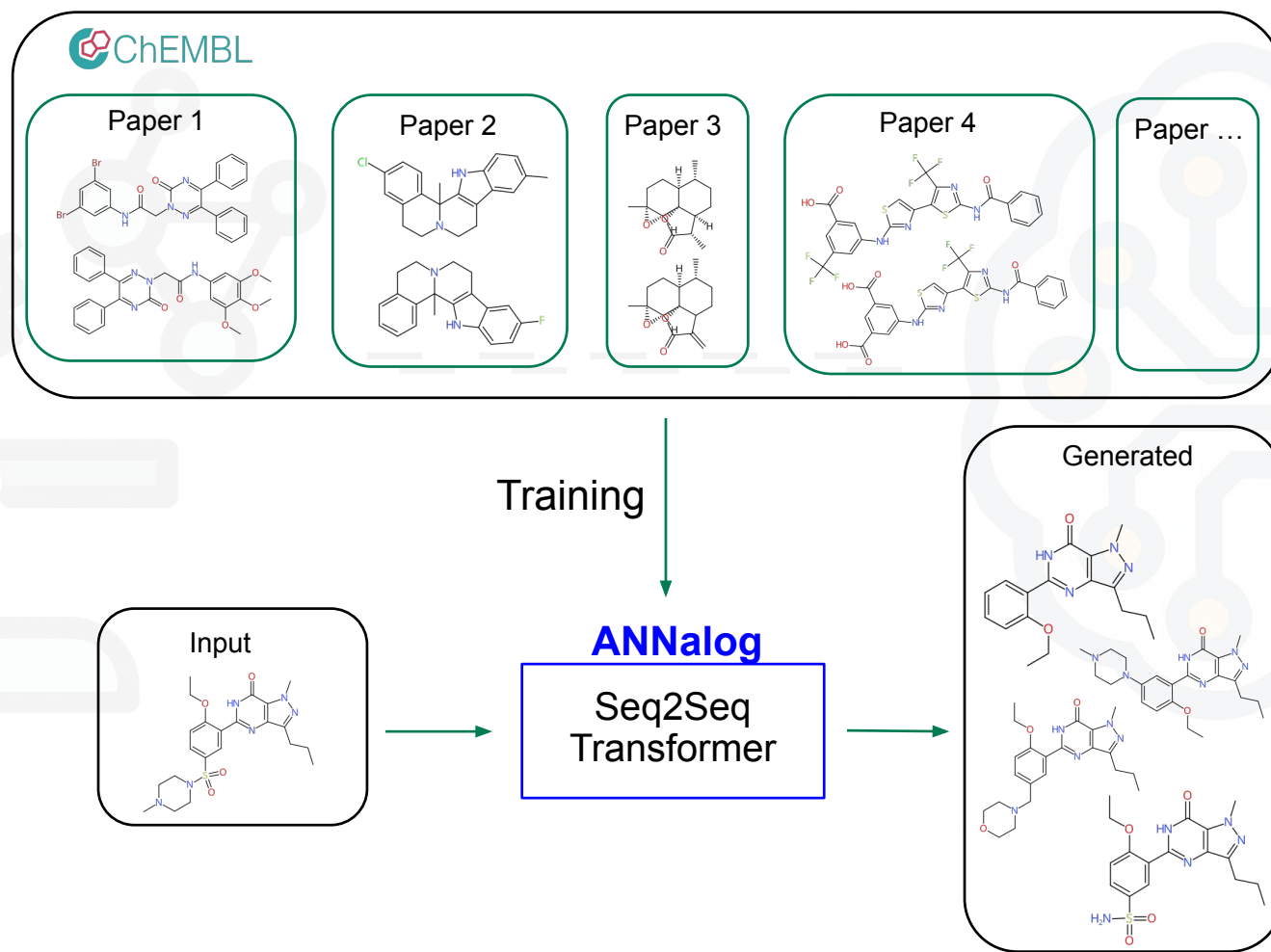
Chemical Biology Resources Team Leader

EMBL-EBI 

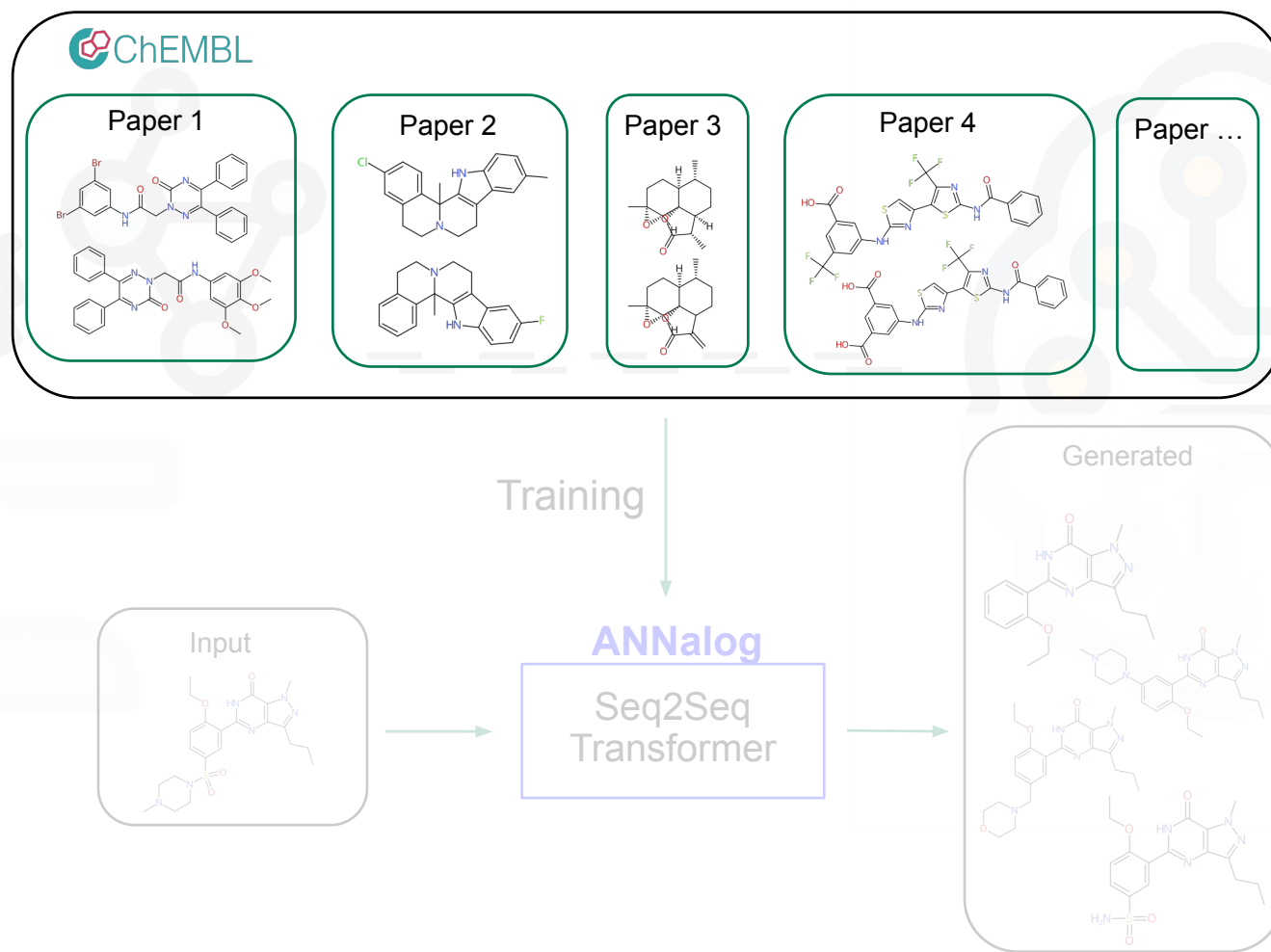
25/08/2026

ACS Fall 2026, Chicago

A generative model for MedChem-similar molecules

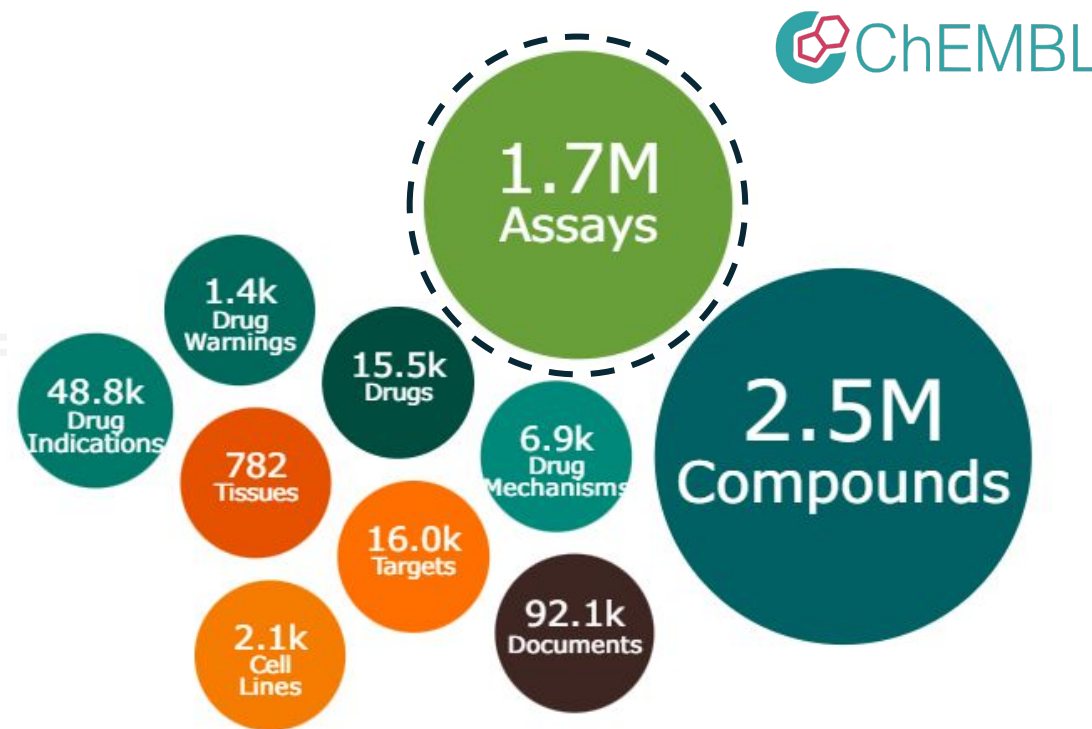


A generative model for MedChem-similar molecules



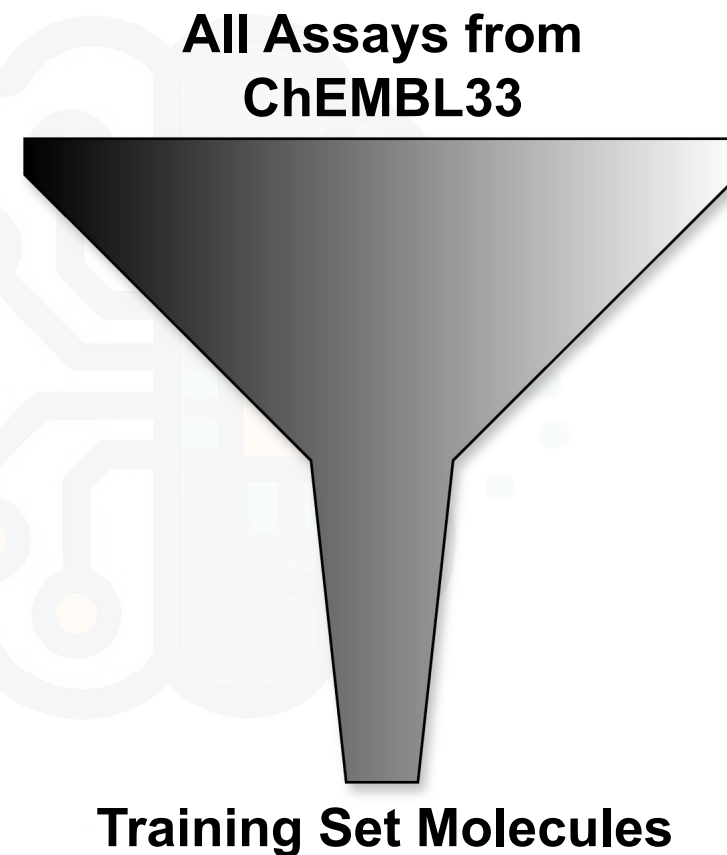
MedChem-similar molecules

- Definition
 - Two molecules can be considered 'similar' if it's plausible for a medicinal chemist to synthesize and test them within the same medicinal chemistry program
- Translate to ChEMBL...
 - We consider molecules from the same assay as MedChem-similar to each other (*)



Training set preparation from ChEMBL

- Binding and Functional assays with $8 \leq N \leq 25$ molecules
- Activity standard type: pIC_{50} , pK_i , pEC_{50}
- Remove:
 - Peptides
 - Large molecules (MW > 600 Da)
 - Molecules containing atoms outside the set {C, N, H, O, S, F, Cl, Br}
 - Molecules with specified isotopes
 - Flexible molecules (> 7 rotatable bonds)
 - 'Popular' molecules (as may not be similar)
 - Drugs/present in Wikipedia/appearing in more than five papers

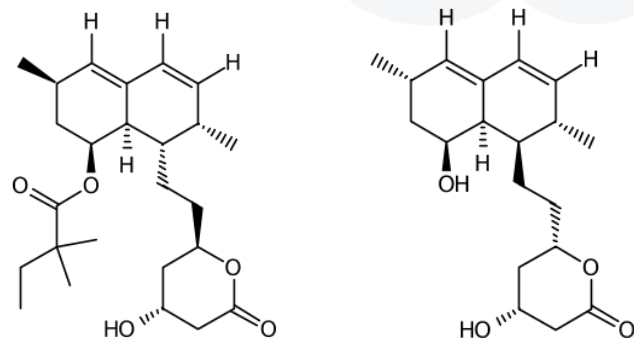


Training set preparation from ChEMBL

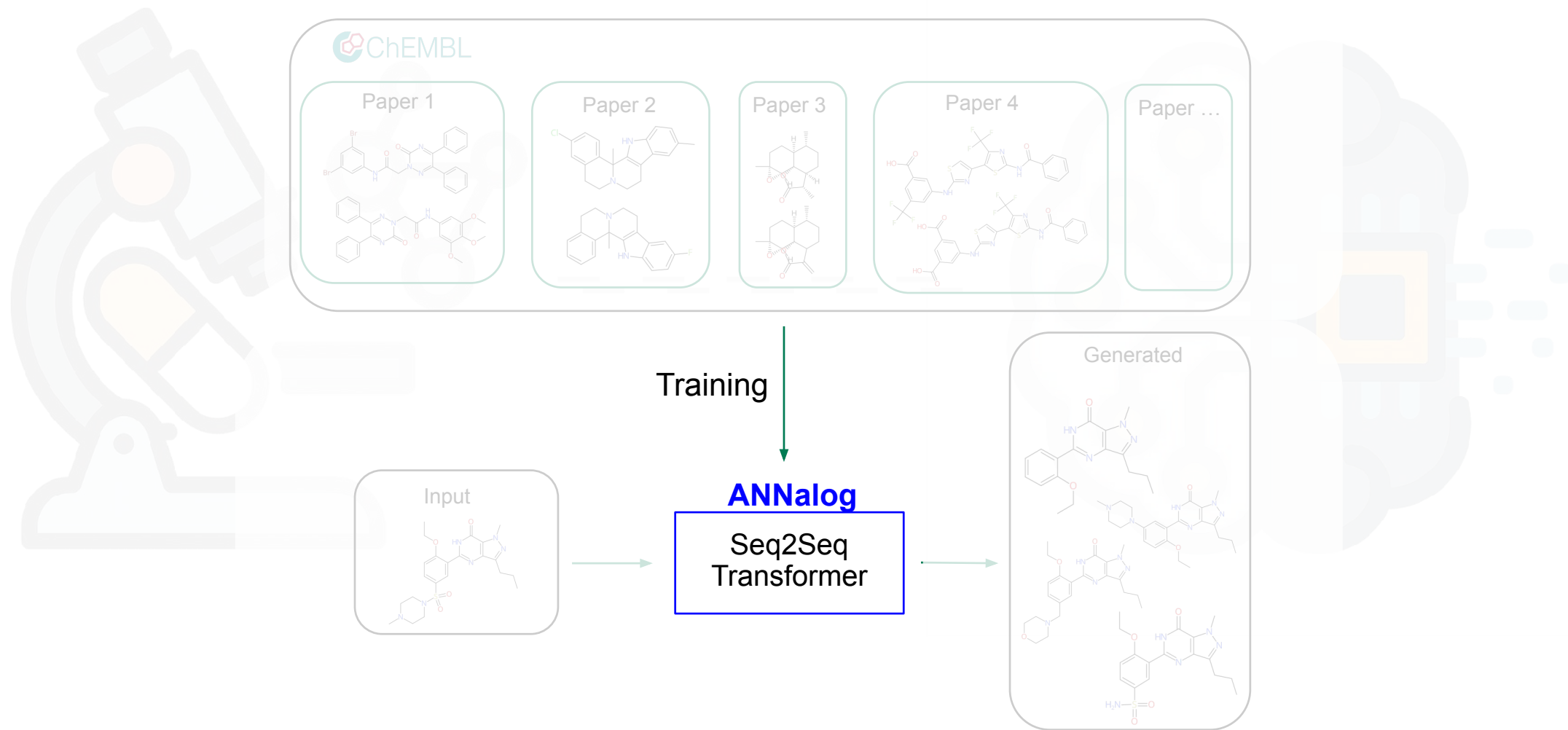
- Generate molecule pairs from same assay
 - Max 30 per paper
 - 608,642 pairs of molecules
- We want ANNalog to preserve existing stereochemical configurations rather than generate trivial permutations
- To reduce the likelihood of ANNalog learning to invert a stereocentre
 - Pairs differing only in stereoconfiguration were dropped
 - Pairs where their scaffolds differed only in stereoconfiguration were dropped

Training Set Molecules

Pairs of MedChem similar molecules

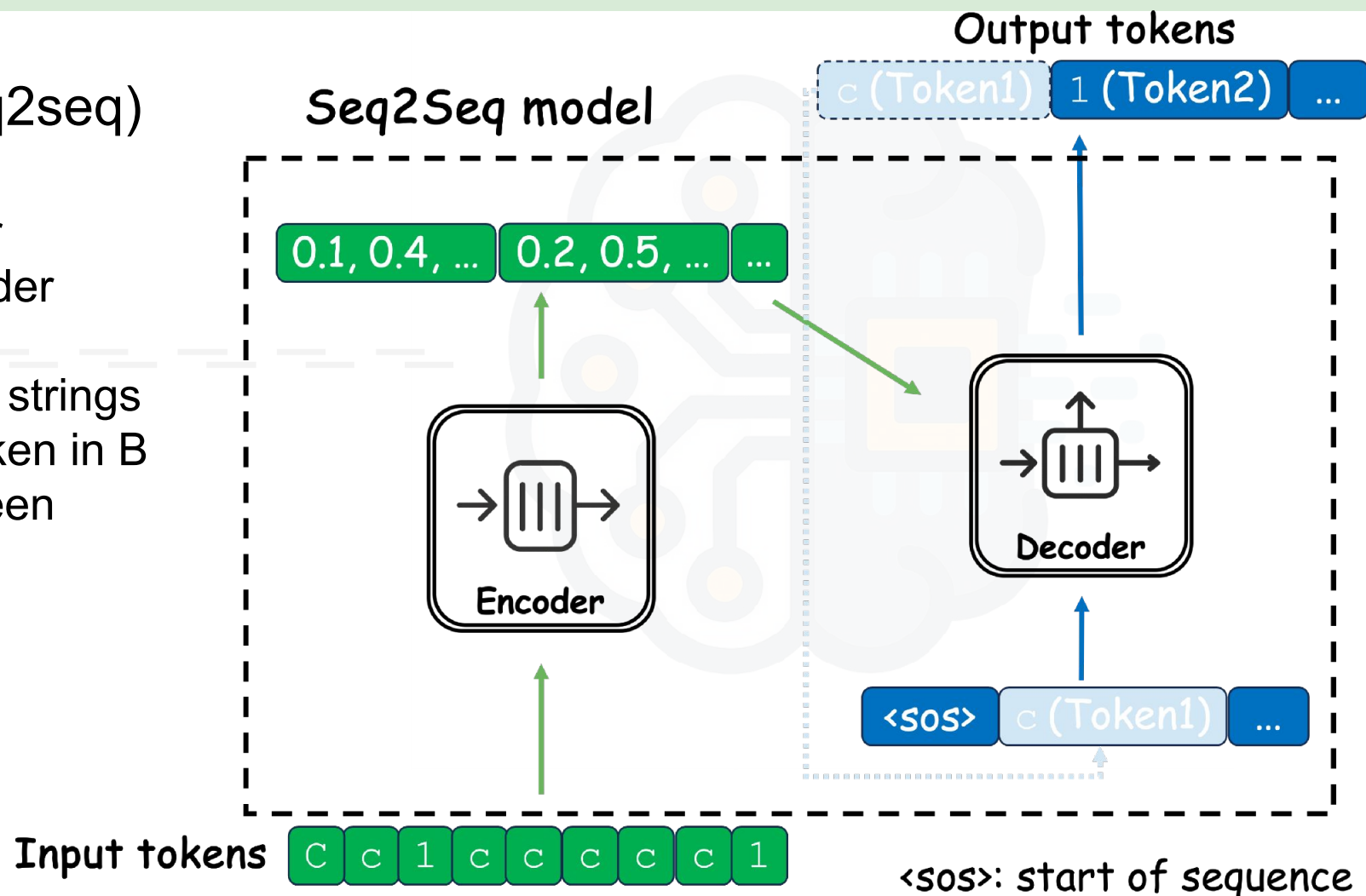


A generative model for MedChem-similar molecules



Model architecture

- Sequence to sequence (seq2seq) model
 - Encoder-decoder transformer
 - 3 encoder layers, 3 decoder layers
 - Trained with pairs of SMILES strings (A, B) to predict each next token in B
 - Learns the correlations between SMILES strings

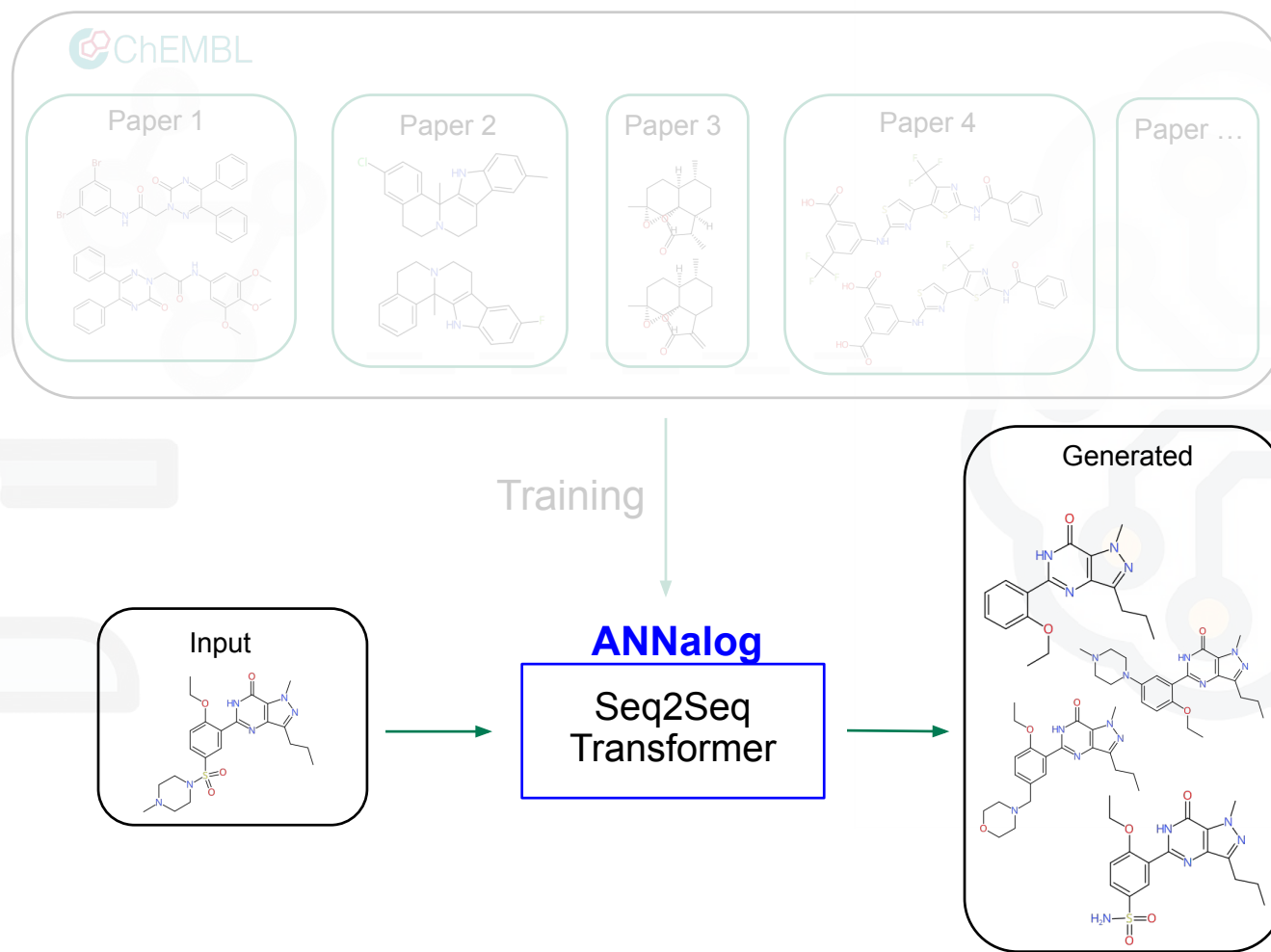


Preparation of pairs of SMILES strings is important

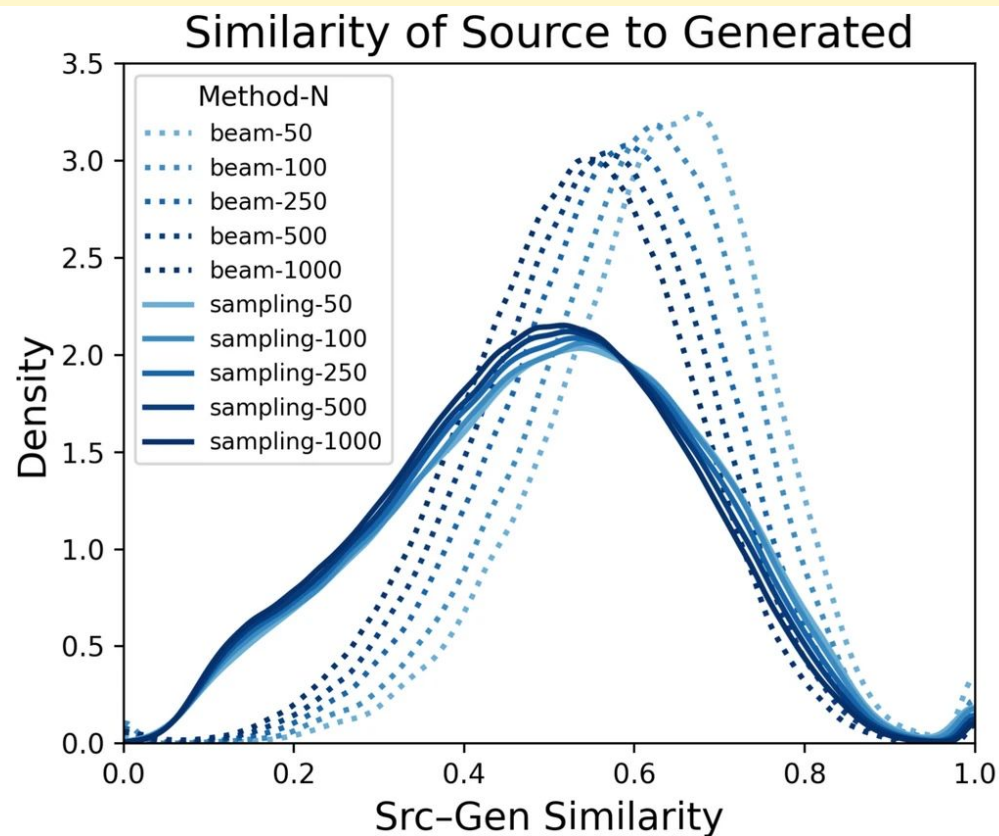
- The seq2seq transformer is trained on pairs of SMILES strings (A, B) in both directions*
 - Given A, suggest B, and given B, suggest A
- The SMILES for the second molecule was aligned to that of the first
 - 10K random SMILES variants were generated
 - The one with the smallest Levenshtein distance was retained
 - This helps ensure that the differences between the SMILES strings correspond to differences between the structures (not differences due to ordering)
- ANNalog makes predictions for a SMILES string as presented (rather than using a canonicalised form)
 - To improve robustness to the presented atom order of the SMILES, five random variants of A are generated (to which B is aligned), and similarly for the reverse order

* One could imagine variants of ANNalog, where (A, B) are presented only in one direction, e.g. ordered by MW or HAC, by complexity, by measured activity, by predicted ease of synthesis, by ring count, by LogP

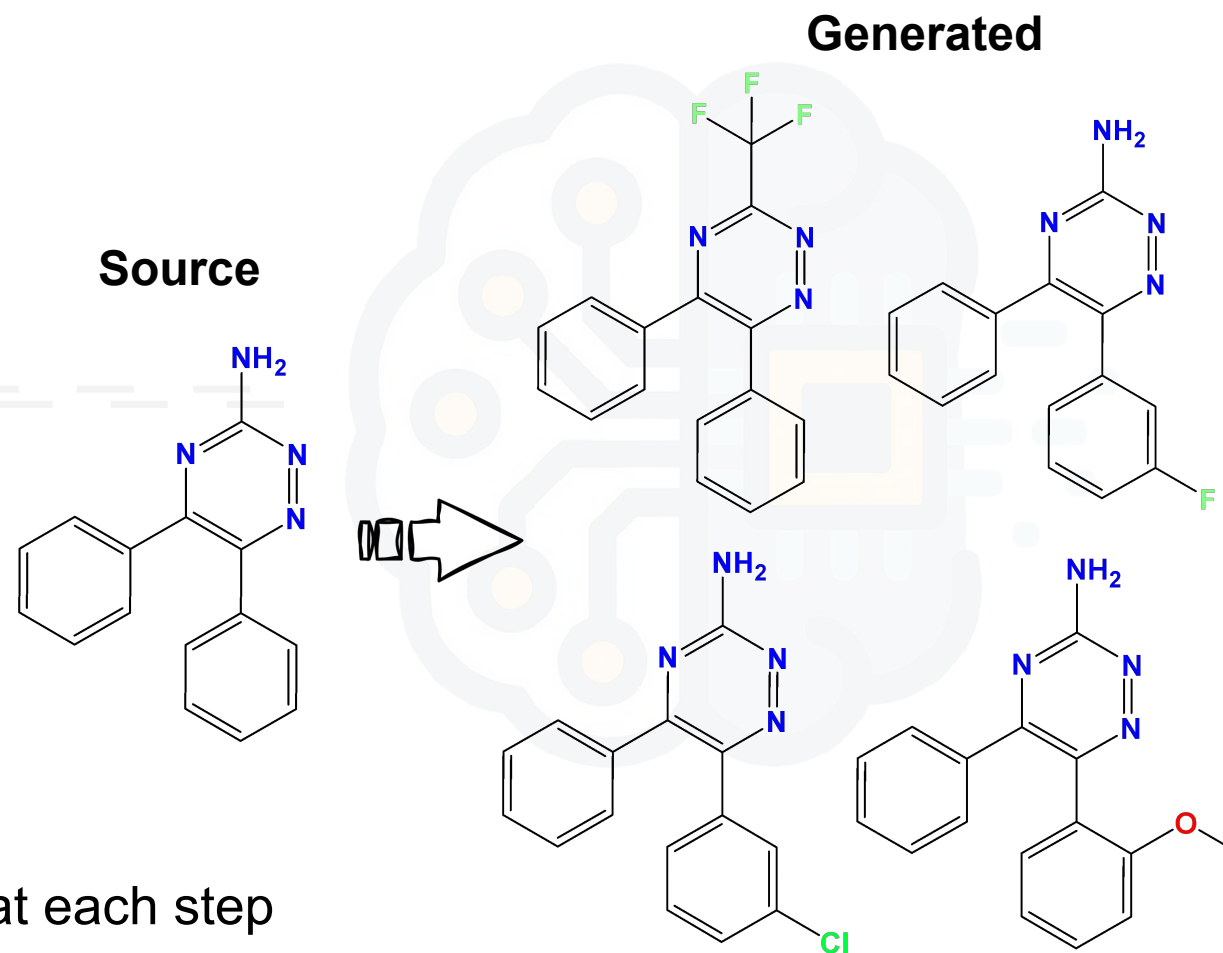
A generative model for MedChem-similar molecules



ANNAlog generates 'fingerprint'-similar molecules

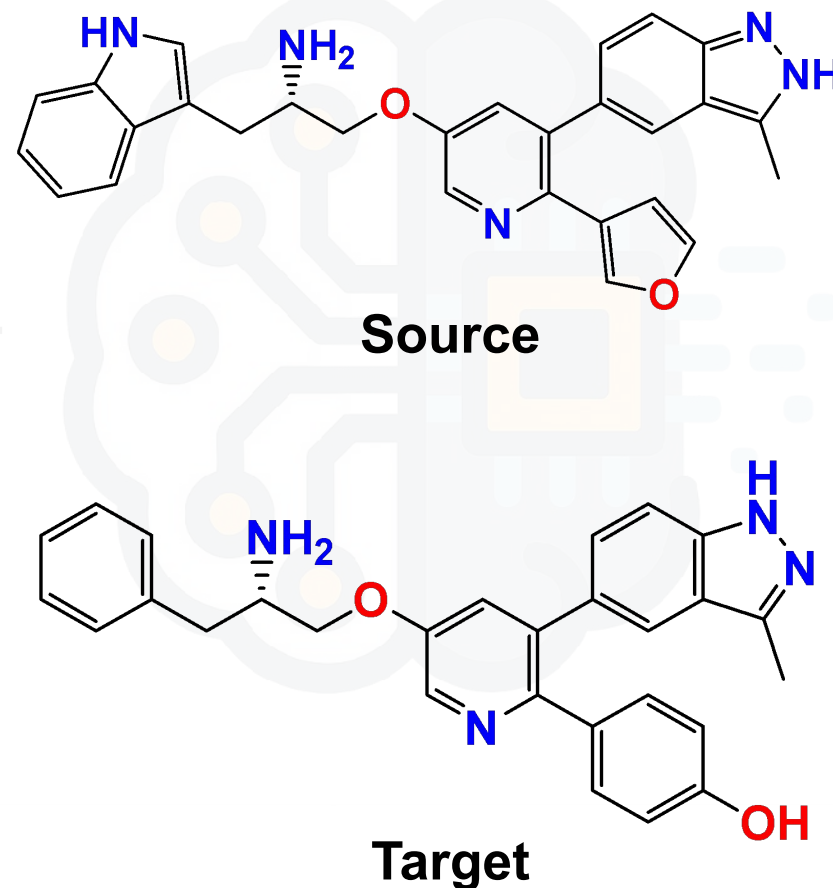


- Beam search
 - Generates the most **probable** solutions at each step
- Sampling (temperature = 1.2)
 - Samples from the distribution of possible tokens



Ability to generate MedChem-similars - dataset

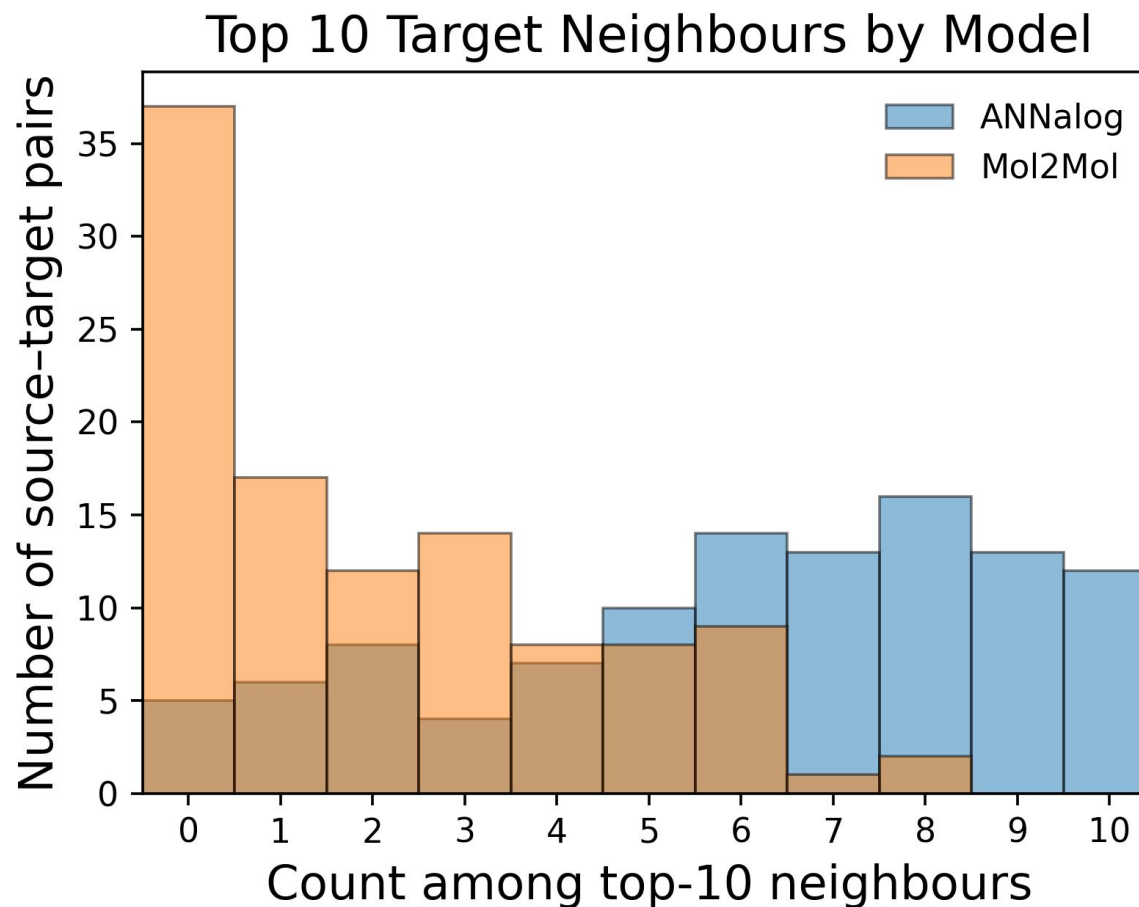
- 108 pairs of molecules (A, B) involving ring system or scaffold transformations were selected manually from additional assays present in ChEMBL35
 - i.e. unseen by the original model trained on ChEMBL33
- For each pair (A, B), ANNalog and Mol2Mol (*) were used to generate 100 molecules (using beam search) given A as input
- The two models were compared based on how many of the top 10 nearest molecules to B came from each



* Tibo A, He J, Janet JP, Nittinger E, Engkvist O. Exhaustive local chemical space exploration using a transformer model. *Nat. Commun.* **2024**, 15, 7315.

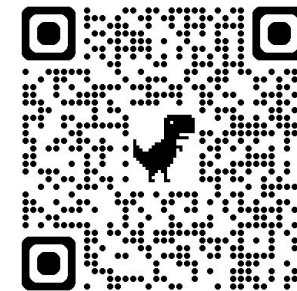
Ability to generate MedChem-similars - results

- Overlapping histograms showing frequency of the number of top 10 neighbours coming from **ANNAlog** or **Mol2Mol**
- On this task, ANNAlog frequently provides the majority of the top 10 neighbours
 - Mol2Mol was developed to explore local chemical space around a molecule



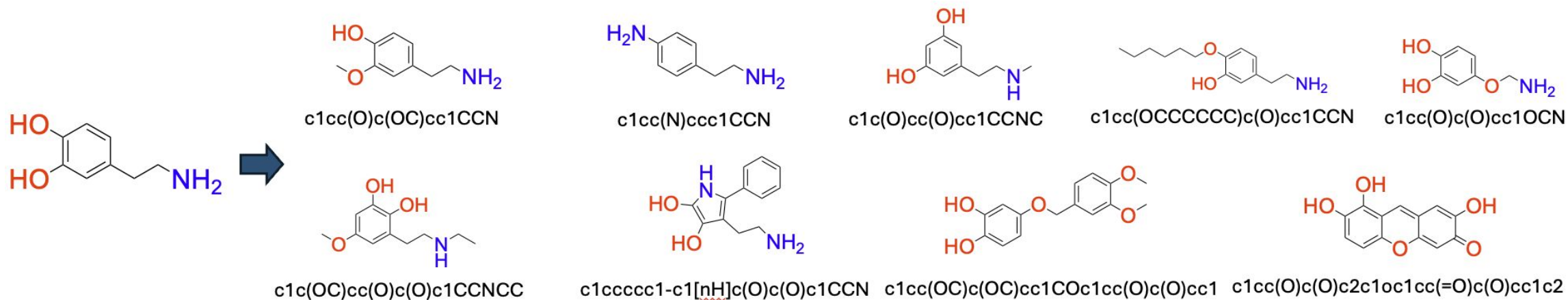
Example of running ANNalog

<https://github.com/DVNecromancer/ANNalog>



```
$ pip install annalog
```

```
$ annalog-generate -i "C1=CC(=C(C=C1CCN)O)O" -m sampling -n 10 -o dopamine.tsv
```



Some practical tips...

- Changes tend to occur on the RHS of the generated SMILES
 - It follows that by presenting the input SMILES string in different ways, the user can selectively focus ANNalog's attention on different parts of the molecule
 - If changes should be distributed evenly, an option “-e variants” generates random SMILES variants and pools the results
- ANNalog allows the user to supply a fixed SMILES prefix to ensure that a particular portion of the molecule is unchanged
- Sampling can be used to create more diverse output
 - Feeding the result through more than once can increase diversity further (see option “-e recursive”)
- Not everything is gold
 - `chembl_gen_check [1]` results are appended to the output to flag unusual structures

Acknowledgements



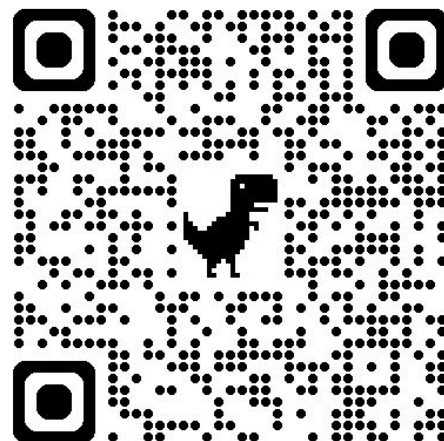
- **Wei Dei**
- Dr. Arianna Fornili



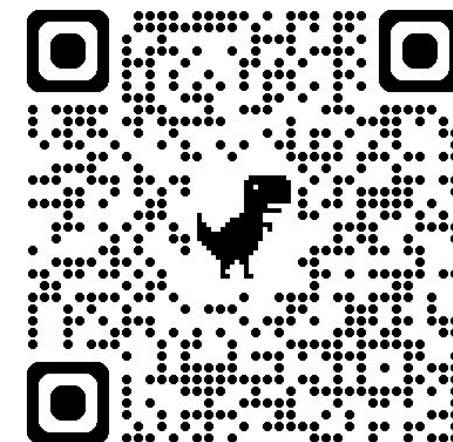
- Dr. Jonathan Tyzack
- Dr. Chris de Graaf



Paper



GitHub



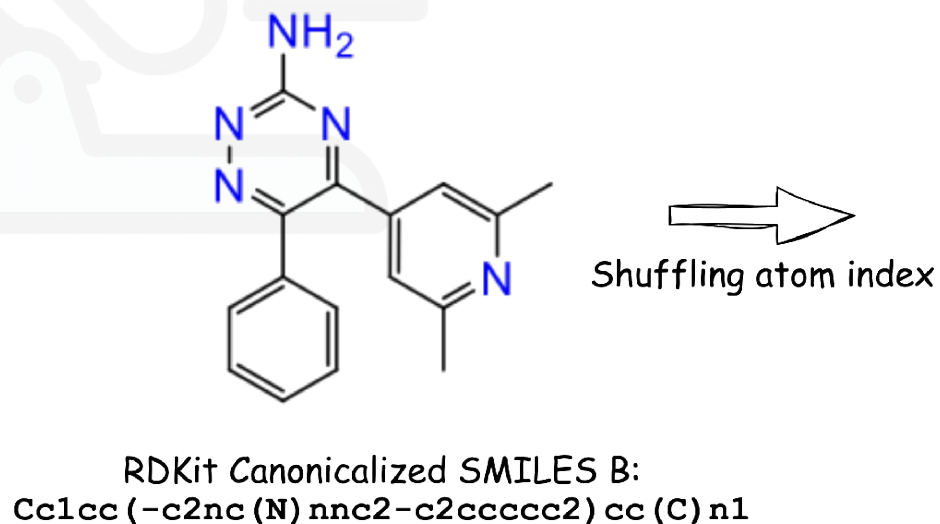
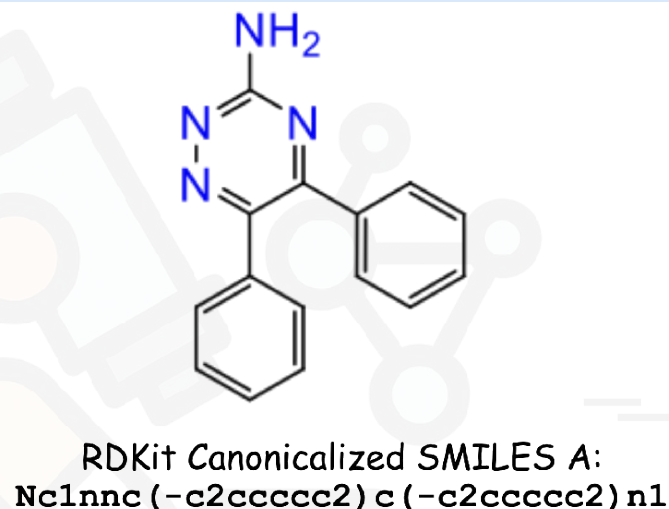
Colab

Conclusion

- ANNalog – a generative model for medchem analogues
 - Training set preparation
 - MedChem molecule pairs from ChEMBL
 - Model training
 - Transformer based Seq2Seq model
 - Generation Analysis
 - ANNalog prefers to generate structurally similar molecules “first”
 - ANNalog has the ability to perform scaffold hopping
 - Proven by MedChem similar pairs
 - Real-world case study

ANNalog – a generative model for medchem analogues

Training Set Preparation – SMILES alignment



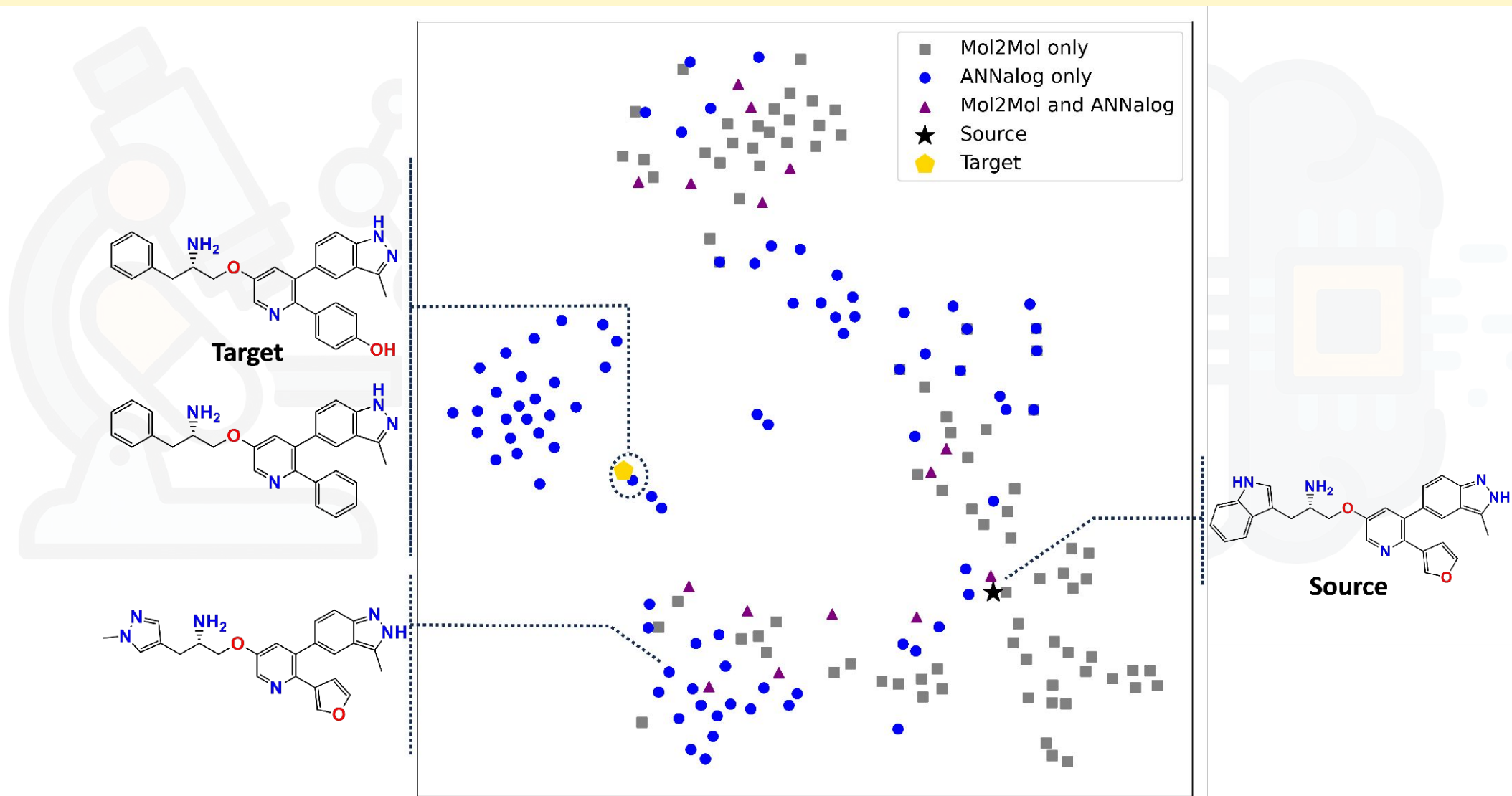
SMILES B' Best aligned with A:
Nc1nnc(-c2ccccc2)c(-c2cc(C)nc(C)c2)n1

String similarity
calculation

1. Nc1nnc(-c2ccccc2)c(-c2cc(C)nc(C)c2)n1
 2. n1c(C)cc(-c2c(-c3ccccc3)nnc(N)n2)cc1C
 3. Cc1nc(C)cc(-c2c(-c3ccccc3)nnc(N)n2)c1
 4. Cc1cc(-c2c(-c3ccccc3)nnc(N)n2)cc(C)n1
 5. c1cc(-c2c(-c3cc(C)nc(C)c3)nc(N)nn2)ccc1
- ⋮

ANNalog – a generative model for medchem analogues

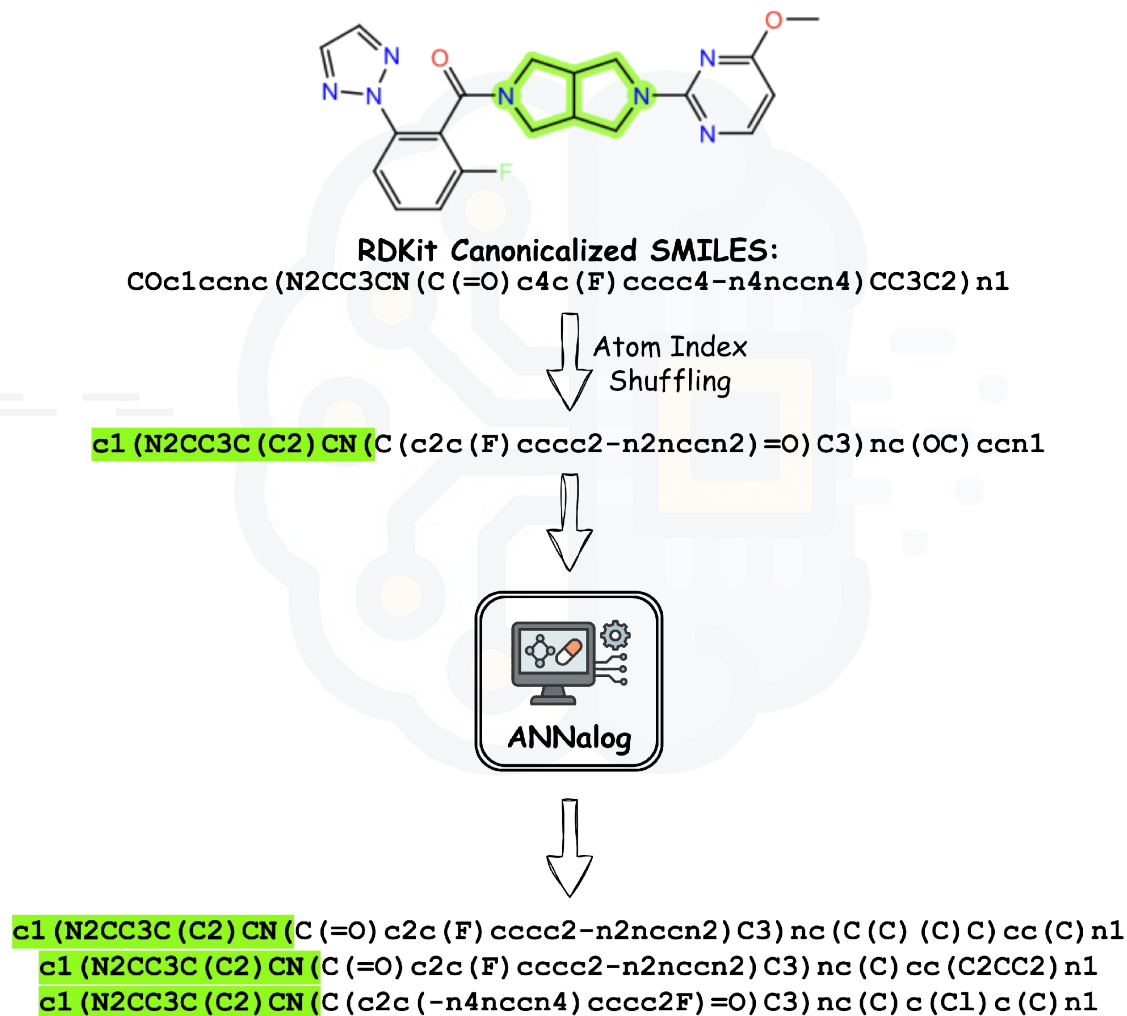
Generation Analysis – Scaffold hopping



ANNalog – a generative model for medchem analogues

Generation Analysis – Scaffold hopping towards constrained direction

- Patent extracted (US20240109901A1)
 - Orexin 2 pyrrolopyrrole antagonists
 - 380 unique, exemplified molecules
 - 121 unique scaffolds
- Generation and analysis
 - Input each molecule within the testing set to the model
 - Generate up to 5000 SMILES per input
 - Standardized input, **prefix-controlled input** and **prefix0 input** (SMILES starting with pyrrolopyrrole structure but not fixing them)



ANNalog – a generative model for medchem analogues

Generation Analysis – Scaffold hopping towards constrained direction

- Scaffold recall
 - How many generated scaffolds re-discovered other scaffolds within the test set
- Sampling decoding method
 - Samples each next token from the predicted probability distribution
- ANNalog could re-discover ~20% scaffolds with prefix control
 - Re-discover ~15% scaffolds without prefix-control (prefix0 input)

